EFU 104.58 URBAN LICENSE v1.0

"The Living City Constitution" – Complete European Edition [conversation context]

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EFU 104.58 Urban License v1.0
"THE CITY'S RIGHT TO ITS OWN METABOLISM"
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Official Designation: EFU 104.58 City License v1.0
Effective: March 1, 2026
Base Document: EFU 104.44 Eternal Life License v1.0
Related: EFU 104.57 Village License v1.0

City vs. Village Paradox

Village: Produces more than consumes (food, water, people, culture).
City: Consumes more than produces (energy, materials, waste, decisions). [conversation context]

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City Metabolism Reality:
INPUT: Food 90-95% external | Water 60-80% distant | Energy 85-95% imported
OUTPUT: Wastewater 100% | Solid waste 95% exported | Urban heat island +2-8°C
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Core Principle: Cities are **metabolically parasitic** – but can be made ethical.

Three Pillars

Pillar	Mechanism	Goal
TRANSPARENCY	Annual EFU Audit	Know exact metabolic debt
RESPONSIBILITY	3-5% annual R-EFU ↓	Measurable reduction targets
JUSTICE	Social Impact Test	No green gentrification

I. CITY METABOLIC AUDIT PROTOCOL

1.1 Mandatory Structure

Annual audit covering 6 flows: Energy | Materials | Water | Waste | Transport | Buildings.

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Total City R-EFU/year = Σ(A+B+C+D+E+F)
R-EFU/person/year = Total / Population
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Benchmarks (2025 estimates):

Category	Population	Examples	R-EFU/person/year
Small City	50k-150k	Győr, Freiburg	80k-120k

Category	Population	Examples	R-EFU/person/year
Medium City	150k-500k	Debrecen, Grenoble	100k-150k
Large City	500k-2M	Budapest, Hamburg	120k-180k
Metropolis	2M+	Berlin, Paris	150k-250k

Visualization: Sankey diagram + Heatmap + 10-year trend [conversation context].

II. SECTOR SUB-LICENSES

104.58-A: URBAN TRANSPORT (-30% R-EFU/fő/10év)

Current (200k city): 45% car → 7,435 R-EFU/person/year.

Target Modal Split: Car 28% | Public 42% | Bike 20%.

Measure	R-EFU Savings	Tools
Tram expansion	-527M/year	15km new lines, e-buses
Bike infrastructure	-1M/year	100km protected lanes
Congestion pricing	Integrated	London model, social rebates

Social Justice: Low-income transit subsidies offset parking fees [conversation context].

104.58-B: URBAN BUILDINGS (-40% R-EFU/m²/20év)

Current: 540k R-EFU/person/year (largest sector!).

Strategy:

- Panel retrofits: 4M m² → -20,000M R-EFU/year
- EFU 104.44 new builds: Passive house + eternal materials
- Demolition mining: Concrete 30% recycled

ROI: €150/m² insulation pays back in 15 years [conversation context].

104.58-C: URBAN FOOD (+5→25% local, -20% R-EFU)

Current: 612M R-EFU/year (meat = 171M!).

Key shifts:

Action	Savings	Method
Urban gardens	-3.6M	50ha community plots
Peri-urban farms	-27.5M	City leases land to farmers
Waste ↓ 30→18%	-73M	Composting + "Too Good"
Meat ↓ 20%	-33.6M	Meatless Monday everywhere

Justice: Vouchers for low-income local food access.

104.58-D: URBAN WASTE (Landfill 70→5%, 15 years)

Current: 65M R-EFU/year (70k t landfill).

Zero Waste Path:

- Bans: Plastic bags (2027), cutlery (2027)
- Composting: 30k t bio → 90% composted (-12.3M)
- Deposit-return: 95% bottles back
- Repair Cafés: 10 sites weekly

104.58-E: URBAN WATER (-20% use, -30% R-EFU)

Current: 7,129M R-EFU/year (150L/person/day).

Targets:

- Leakage 25→10% (-657M)
- Rainwater harvest: 500k m^3/year (-200M)
- Decentralized treatment: 15% on-site (-527M)

III. URBAN GOVERNANCE

EFU Urban Council (15 members)

Fixed: Mayor | Sustainability Chief | Planner | Transport | Utilities
Elected: Architect | Engineer | Economist | District reps (5-6)
Advisors: NGO | Youth | Seniors

Duties:

1. Approve annual metabolic audit
2. Set 5-year R-EFU reduction plan
3. Social impact veto power
4. Public dashboard + press conference [conversation context]